

The Same Curve, Six Times

Surnames, counties, campaign money and readers, sorted largest to smallest – and how much of a distribution you have to throw away before it becomes a law

84-355 Democracy's Data

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Six quantities, taken from six chapters of this book that have nothing to do with each other. How many Americans share a surname. How many people live in a county. How much money a candidate raised, how much a committee spent independently, and how much was spent on a candidate by other people. And how many readers opened one Wikipedia article on one day.

Different agencies, different decades, different units: people, dollars, clicks. 168,595 numbers in all.

Sort each one from largest to smallest and plot it against its rank, and they are the same picture.

That is the observation this chapter starts from, and it is genuinely surprising. What the chapter is actually about is the sentence people say next: *it's a power law*. That sentence turns out to depend almost entirely on how much of the data you are willing to leave out.

01 One row is a rank

Every row of `rank.csv` is one thing and its position in a sorted list.

Series	Rank	Value	What it is
Surnames	1	2,369,644	SMITH
Surnames	10	1,039,848	MARTINEZ
Counties	1	10,014,009	Los Angeles, California
Counties	10	2,418,185	Riverside, California

Column	What it holds	Measurement
series	which of the six quantities this row belongs to	categorical
rank	position in that quantity's sorted list; 1 is the largest	count
value	the quantity itself – people, dollars or views	continuous
who	the surname, county, candidate, committee or article-day it belongs to	identifier

The file is thinned and the thinning is not a sample. There are 168,595 values behind these figures and 1,276 of them are drawn. On a logarithmic axis ranks 100,000 and

100,001 land on the same pixel, so every one of the first fifty ranks is kept and then points at evenly spaced *logarithmic* intervals after that. Every summary number in this chapter, every mean and every fitted slope, is computed from all 168,595 values before the thinning. Only the drawing is thinned.

One row was removed before any of this, and one whole series had to be refetched.

The Census surname file's largest row is not a surname. The Bureau publishes every name held by 100 or more people, and sweeps everyone else into a single line called *ALL OTHER NAMES*. That is 36,257,637 people, 12.1% of the country. Left in, it would have outranked SMITH twelve times over and sat at the top left of Figure 1 as the most important-looking point in the chapter. It is a residual, the category for what is left over rather than a name, and the file flags it by giving it rank 0.

The pageview series was wrong, and the bottom of this chapter's curve is what found it. English Wikipedia's article on JD Vance sat under a different title until it was renamed in late July 2024, and the Pageviews API reports traffic **by title**. So a request for the title the article has today returned, for every day before the rename, only the people who arrived through the leftover redirect. A sitting United States senator appeared to have a handful of readers a day through the spring, and about 190 of those days sat at the very bottom of the smallest series here. The media-attention chapter now requests every title the article has held and sums them, and this chapter reads the corrected file rather than working around it.

That is the general hazard, and it is worth stating before any of the figures: **a long tail is where measurement error goes to hide**. A value that is three orders of magnitude too small is conspicuous at the top of a ranking and invisible at the bottom, where everything is small anyway.

Before you read on, commit to a number. SMITH is the most common surname in the United States, held by 2,369,644 people. **How many people do you think hold the 100th most common surname?** Write the number down before you scroll.

02 Six curves

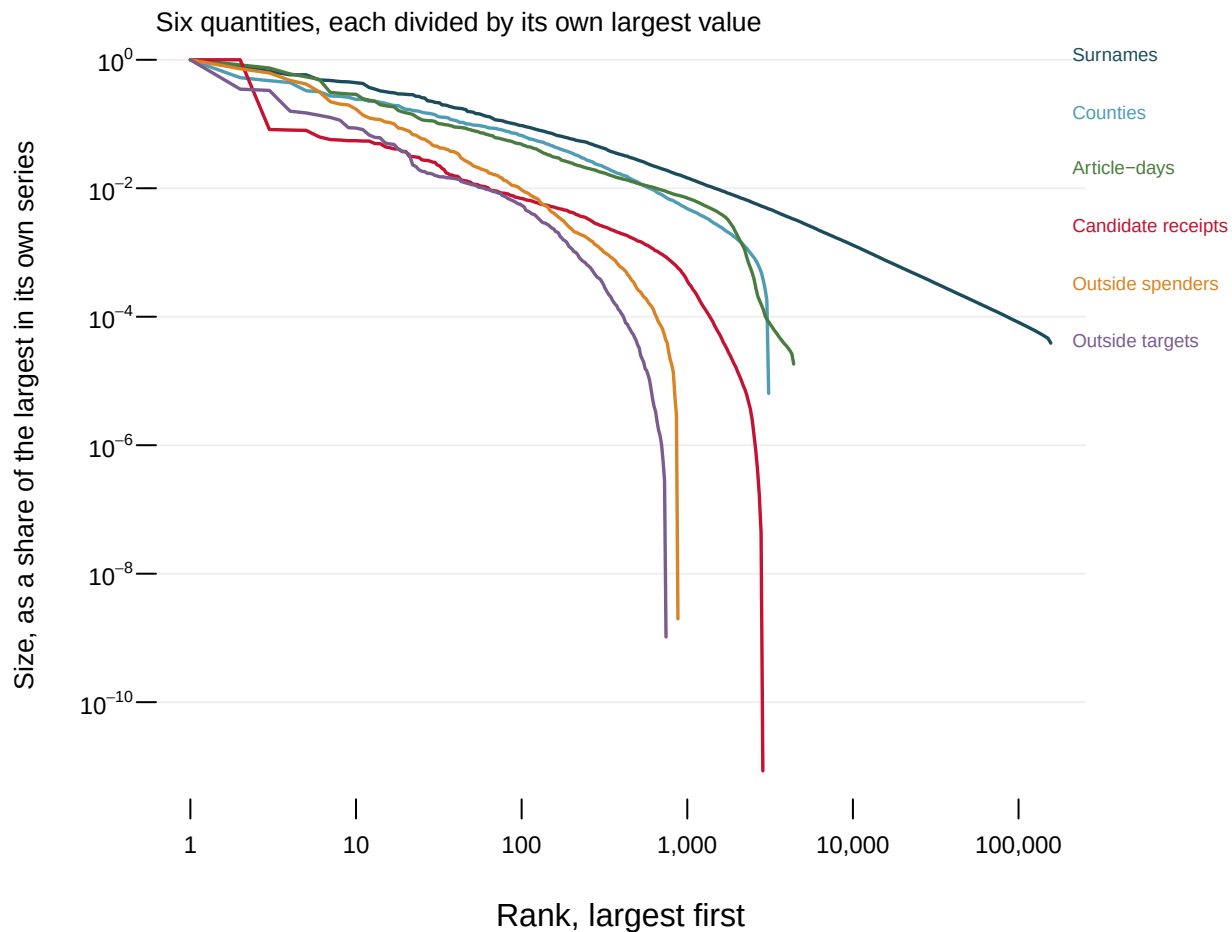


Figure 1. Six unrelated quantities, each sorted largest to smallest. Both axes are logarithmic. Each series is divided by its own largest value. On log axes that shifts a curve up or down without changing its slope, so all six start at the top left, and the only thing being compared is how fast each one falls. On screen, hover any point to see what it is, and the button redraws the same numbers on ordinary axes.

Try that button. On ordinary axes all six collapse into the same L: a vertical cliff at the left edge and a flat line the rest of the way. That is not a failure of the drawing. It is the honest straight-axis picture of a quantity whose largest value is millions of times the median, and it is what a spreadsheet's default chart will give you every time. **The logarithms are not a cosmetic choice. They are the only reason there is anything to look at.**

With them, six curves that had no reason to resemble each other are close to parallel over the first two or three decades. Los Angeles, California has 10,014,009 people and SMITH covers 2,369,644; the shapes of the two lists are nearly the same shape.

03 Answer to the prediction

Series	Rank	What is there	Observed	Zipf predicts	Ratio
Surnames	1	SMITH	2,369,644	2,369,644	1.00
Surnames	10	MARTINEZ	1,039,848	236,964	4.39
Surnames	100	LONG	223,485	23,696	9.43
Surnames	1,000	COMPTON	34,287	2,370	14.47
Candidate receipts	1	HARRIS, KAMALA	\$1,175,189,365	\$1,175,189,365	1.00
Candidate receipts	10	GALLEGO, RUBEN	\$64,657,200	\$117,518,937	0.55
Candidate receipts	100	RYAN, PATRICK	\$8,143,126	\$11,751,894	0.69
Candidate receipts	1,000	MONROE, KATE M	\$424,374	\$1,175,189	0.36

The oldest and simplest version of the claim is **Zipf's rule** (Zipf 1949): the item at rank k is $1/k$ of the item at rank 1. The 10th most common surname should be a tenth of SMITH; the 100th, a hundredth.

It is not close. The 100th most common surname, LONG, is held by 223,485 people – 9.4 times what the rule predicts. Most people guess too low here, and the reason is that the rule is too steep near the top: surnames fall away more gently than $1/k$.

Money misses in the other direction. The 100th best-funded candidate raised \$8,143,126, which is 0.69 of the prediction – the drop below the top handful is steeper than $1/k$, not gentler. **Zipf's rule fails on both of these, and it fails in opposite directions.** Any single number offered as *the* exponent is going to have the same problem.

04 The mean is not the middle

Series	Things	Median	Mean	Mean over median	Above the mean	Top 1% holds
Surnames	156,621	264	1,677	6.3	10.6%	51.1%
Counties	3,100	25,774	105,050	4.1	18.4%	24.1%
Article-days	4,392	6,066	36,544	6.0	22.5%	31.6%
Candidate receipts	2,860	\$95,610	\$2,393,764	25.0	13.4%	55.6%
Outside spenders	878	\$225,415	\$4,745,612	21.1	11.4%	50.7%
Outside targets	744	\$144,344	\$5,600,333	38.8	12.8%	52.2%

Before any curve is fitted, the table above is the practical consequence, and it is the part of this chapter that will change how you read a news story.

In every one of these six, **the mean describes almost nobody**. The typical candidate in the FEC file raised \$95,610; the *average* candidate raised \$2,393,764, and only 13.4% of them are above it. For outside money spent on a candidate the mean is 38.8 times the median, and 95 of 744 candidates clear it, or 12.8%. Across all six series the share of things above their own mean runs from 10.6% to 22.5%.

A quantity like this has no typical value in the way height or blood pressure does. “The average PAC spent five million dollars” is arithmetically true and descriptively false: it is a statement about HARRIS, KAMALA, divided by 744 and handed to everyone else.

The top 1% holds between 24.1% and 55.6% of each total. Gini coefficients run 0.760 to 0.923 – every one of them above the household-income Gini of any country on earth.

05 How straight is straight

Now the claim itself. If these are power laws, then $\log(\text{size})$ against $\log(\text{rank})$ is a straight line, and fitting one is a two-line exercise that any software will do without complaint. The question nobody asks out loud is **which part of the data the line was fitted to**.

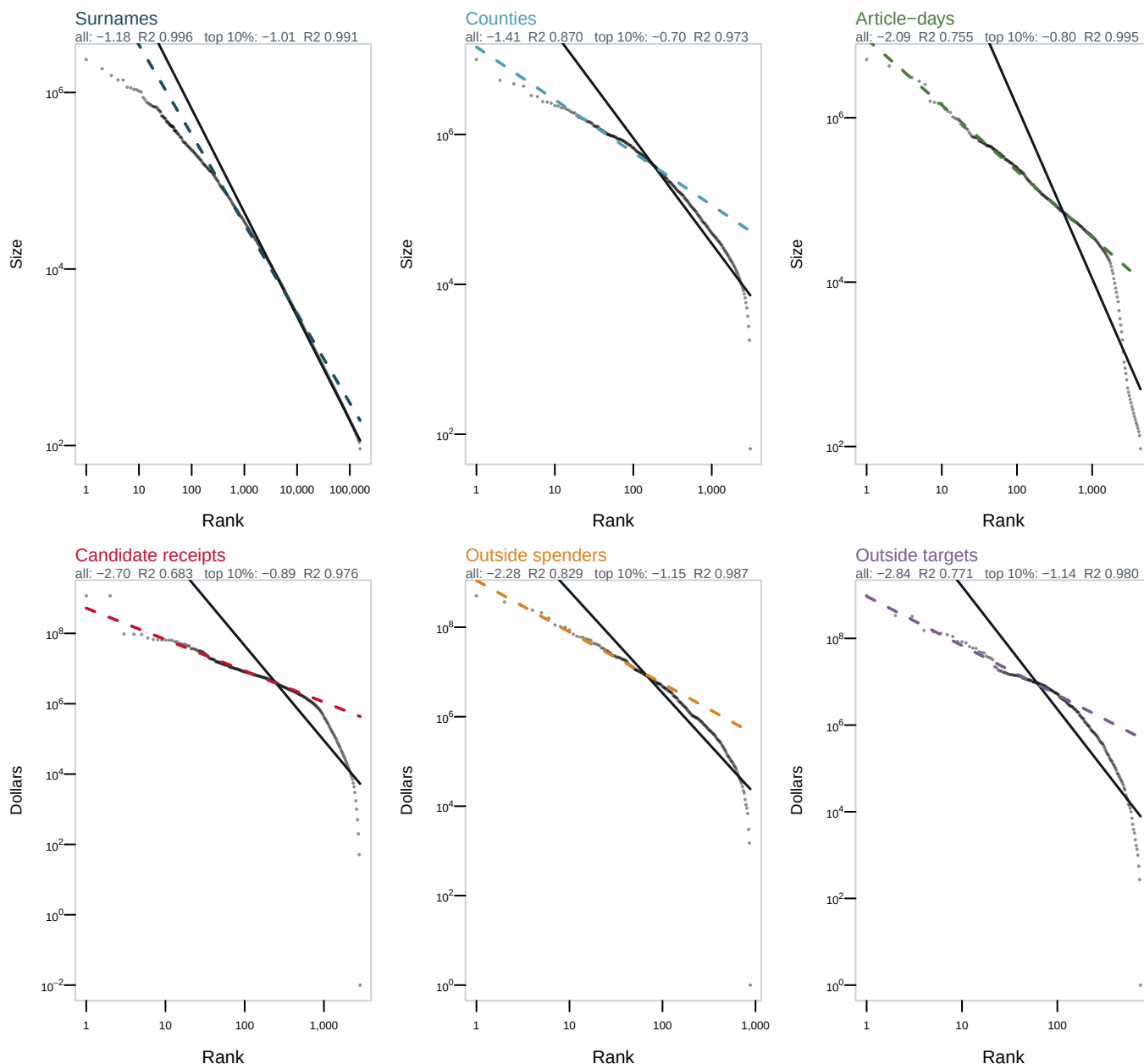


Figure 2. The same fit, at forty-four different cut-offs. On screen: pick a series with the buttons, and the slider sets how much of it the straight line is fitted to, from the top 1% to all of it. The heavy segment covers the fitted range, the grey dashed extension is the same line carried out beyond the data it was fitted to, and faded points are outside the fit. The two panels along the bottom show the slope and R^2 at every cut-off at once, with the current one marked. In print, all six series are shown at two fixed cut-offs — the black line is fitted to everything, the coloured dashed line to the top 10%. Either way the fits are least squares on all 168,595 values, not on the points drawn.

Take **candidate receipts**, which the slider walks through. Fitted to the whole file, the line has slope -2.70 and an R^2 of 0.683 — visibly wrong, bending away from the data at both ends. Fitted to the top 10%, the same data gives slope -0.89 and R^2 0.976 . Nothing about the file changed. What changed is that 986 of 2,860 candidates raised nothing at all and are already excluded, and the 2,574-odd who raised something small were, on the second fit, excluded too.

That pattern holds for all six:

Series	Slope, top 10%	R^2 , top 10%	Slope, all of it	R^2 , all of it
Surnames	-1.01	0.991	-1.18	0.996
Counties	-0.70	0.973	-1.41	0.870
Article-days	-0.80	0.995	-2.09	0.755
Candidate receipts	-0.89	0.976	-2.70	0.683
Outside spenders	-1.15	0.987	-2.28	0.829
Outside targets	-1.14	0.980	-2.84	0.771

Over the full range, exactly one of the six is straight: people sharing a surname, R^2 0.996 . The rest run down to 0.683 . Fit only the top decile and **every one of them is a clean power law** — R^2 at worst 0.973 , and all six slopes inside the narrow band from -1.15 to -0.70 .

This is the whole trick, and it is not usually a conscious one. A power law is a claim about the *tail*, so it is entirely normal to fit only the tail. But “only the tail” is a free choice, and the smaller you make it the better any smooth falling curve will fit a straight line. Every curve is straight if you look at a short enough piece of it. **A high R^2 from a tail-only fit is not evidence of a power law. It is evidence that you chose a short tail.**

The slope panel at the bottom of Figure 2 is where to look, because a genuine power law has *one* exponent: move the cut-off and the slope should stay put. None of these six manages that. The steadiest, people sharing a surname, drifts from -0.80 to -1.18 , a spread of 0.38 . It drifts smoothly, in one direction, which is what a lognormal does and a power law does not.

Money a candidate raised slides from -0.94 to -2.70 , a spread of 1.75 . The exponent nearly triples depending on where you stop. There is no number to report there. There is a curve that gets steeper the further down you look.

A second prediction, before the last section. Suppose the FEC changed its reporting threshold so that candidates raising under \$5,000 no longer had to file. Nothing about anyone’s fundraising would change. **What would happen to the slope fitted to the whole file, and what would happen to the slope fitted to the top 10%?** Say which direction each moves and why.

06 What is actually going on

Three ordinary mechanisms produce curves like these, and none of them is exotic.

Multiplying, rather than adding. Suppose a thing's growth in each period is a random *percentage* of its current size, rather than a random amount. Then the sizes after many periods are lognormal: heavy-tailed, straight-ish over a stretch of a log-log plot, and curved over the whole of it. Counties grow this way, and so does a name passed down a family tree. This is the most common explanation, and the one most often misreported as a power law. A lognormal and a power law are hard to tell apart on exactly the range people plot.

Rich-get-richer. When the chance of getting the next unit is proportional to how many you already have, you get a genuine power law. Campaign money has an obvious version: donors give to candidates who look viable, and looking viable means having raised money.

A floor and a ceiling written into the file. The FEC has reporting thresholds; the Census suppresses names under 100 people; a county cannot have fewer than zero residents. The bottom of every one of these curves is partly a fact about administration rather than about the world. That is exactly why the slider matters: the fit that includes the most data is also the fit most contaminated by the rules of the filing system.

07 What this chapter cannot tell you

A straight line on a log-log plot is not proof of a power law, and this chapter deliberately does not test one. Distinguishing a power law from a lognormal takes maximum-likelihood estimation with a fitted lower cut-off and a goodness-of-fit test against synthetic data — the Clauset, Shalizi and Newman procedure in the sources. Least squares on log-log points, which is what Figure 2 does and what most published claims do, is known to be biased and has no standard error worth quoting. Figure 2 is here to show that the answer moves when you move the cut-off, not to supply a defensible exponent.

R^2 is doing very little work here. On log axes it is computed over data spanning many orders of magnitude, and almost any decreasing curve scores well. An R^2 of 0.973 in the fit table above should not persuade you of anything; the point of printing it is that the *same file* also produces 0.683.

These six lists are not the same kind of object. Surnames and counties are close to complete enumerations. Candidate receipts, committee spending and its targets exist only because a filing rule requires them, so their bottom ends are set by that rule. And the very bottom is often not a small participant but a strange one. The smallest total in the receipts file is \$0.01, filed by PARK, CHARLES, and the smallest independent-expenditure committee spent \$1.

Pageviews are a machine's count of requests, not a count of readers. Putting them on one figure compares *shapes*, and nothing else.

One row was removed and one series was fetched again, as described in the second section. The row is the Census's ALL OTHER NAMES bucket, holding 36,257,637 people. The series is the whole Wikipedia pageview file, which had been captured under the wrong title for one of its twelve articles. The removal makes the figures better-behaved, which is precisely why it is stated here rather than buried in the preparation. The re-fetch

is a reminder that this chapter's own subject, the shape of the bottom of a distribution, is also the place a bad measurement is hardest to see.

Concentration is not the same as unfairness, and this chapter measures the first. That 55.6% of campaign money sits with 1% of candidates is a description. Whether that is a problem depends on arguments about representation that the campaign-finance and independent-expenditures chapters take up, and that no exponent settles.

08 Sources

- U.S. Census Bureau. *Frequently Occurring Surnames from the 2010 Census*. Captured and committed by this corpus's surnames chapter; names held by fewer than 100 people are not published individually, and the residual row is dropped here. <https://www2.census.gov/topics/genealogy/2010surnames/names.zip>
- U.S. Census Bureau, 2020 Decennial Census, county population, as committed by this corpus's mapping chapter.
- Federal Election Commission. *Candidate summary file, 2023-2024 cycle, and independent expenditures*, as committed by this corpus's campaign-finance and finance-network chapters. <https://www.fec.gov/data/browse-data/?tab=bulk-data>
- Wikimedia Foundation. *Pageviews API*, English Wikipedia, daily, 2024, as committed by this corpus's media-attention chapter. https://wikimedia.org/api/rest_v1/#/Pageviews%20data
- Zipf, George K. (1949). *Human Behavior and the Principle of Least Effort*. Addison-Wesley. The origin of the $1/k$ rule checked above.
- Clauset, Aaron, Cosma Rohilla Shalizi and M. E. J. Newman (2009). "Power-Law Distributions in Empirical Data." *SIAM Review* 51(4): 661-703. The paper that showed how many published power laws do not survive a proper test, and supplies the procedure this chapter declines to shortcut.
- Gabaix, Xavier (2016). "Power Laws in Economics: An Introduction." *Journal of Economic Perspectives* 30(1): 185-206. On the multiplicative and rich-get-richer mechanisms in the section above.
- The six committed files are read, sorted, thinned for drawing, and every cut-off fitted. It runs in seconds and opens no network connection.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`fits.csv`, `ranks.csv`, `summary.csv`, `zipf.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `summary.csv` has `mean_over_median` and `above_mean_pct` for six very different series. Compare them. What do surnames, counties, money and readers have in common?
- `fits.csv` fits the same curve to different fractions of each series, with `r2`. Follow `r2` as `frac` grows. How much of a distribution do you have to discard before it looks like a law?

- `zipf.csv` puts the observed value beside what the rule predicts, with a `ratio`. Find where the rule breaks down for each series. Is it the top or the tail?
- `summary.csv` reports `top1_share` for each series. Which of the six is most concentrated? Then decide whether the same explanation could possibly cover all six.